

Cálculo de $\mathbb{E}[G_t \mid S_t = s]$ por marginalización del modelo gráfico del MDP

1 Convención de Sutton & Barto

Variables aleatorias en orden temporal:

$$S_t, A_t, R_{t+1}, S_{t+1}, A_{t+1}, R_{t+2}, S_{t+2}, A_{t+2}, R_{t+3}, \dots$$

con dependencias

$$A_{t+j} \sim \pi(\cdot \mid S_{t+j}), \quad S_{t+j+1} \sim p(\cdot \mid S_{t+j}, A_{t+j}), \quad R_{t+j+1} \sim p(\cdot \mid S_{t+j}, A_{t+j}, S_{t+j+1}).$$

El retorno es

$$G_t = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1}.$$

Condicionamos en $S_t = s$.

2 La esperanza como suma contra la conjunta

$$\mathbb{E}[G_t \mid S_t = s] = \sum_{r_{t+1}, r_{t+2}, \dots} \left(\sum_{k=0}^{\infty} \gamma^k r_{t+k+1} \right) p(R_{t+1} = r_{t+1}, R_{t+2} = r_{t+2}, \dots \mid S_t = s).$$

3 Reagrupar por índice de recompensa

$$\begin{aligned} \mathbb{E}[G_t \mid S_t = s] &= \sum_{k=0}^{\infty} \gamma^k \sum_{r_{t+1}, r_{t+2}, \dots} r_{t+k+1} \\ &\quad \times p(R_{t+1} = r_{t+1}, R_{t+2} = r_{t+2}, \dots \mid S_t = s). \end{aligned}$$

Para cada k fijo, sumar la conjunta sobre todas las recompensas excepto R_{t+k+1} da la marginal $p(R_{t+k+1} = r \mid S_t = s)$:

$$\mathbb{E}[G_t \mid S_t = s] = \sum_{k=0}^{\infty} \gamma^k \sum_r r p(R_{t+k+1} = r \mid S_t = s).$$

4 Marginalizar cada recompensa individualmente

Fijamos k . La recompensa R_{t+k+1} depende de $(S_{t+k}, A_{t+k}, S_{t+k+1})$. La conjunta sobre las variables relevantes, dado $S_t = s$, es

$$\begin{aligned} p(A_t, S_{t+1}, R_{t+1}, \dots, S_{t+k}, A_{t+k}, S_{t+k+1}, R_{t+k+1} \mid S_t = s) \\ = \pi(A_t \mid s) p(S_{t+1} \mid s, A_t) p(R_{t+1} \mid s, A_t, S_{t+1}) \dots \\ \times \pi(A_{t+k} \mid S_{t+k}) p(S_{t+k+1} \mid S_{t+k}, A_{t+k}) p(R_{t+k+1} \mid S_{t+k}, A_{t+k}, S_{t+k+1}). \end{aligned}$$

Sumamos sobre todas las variables excepto R_{t+k+1} . Ocurren dos cosas:

1. **Las recompensas intermedias se eliminan.** Para cada $j = 0, \dots, k-1$,

$$\sum_{r_{t+j+1}} p(R_{t+j+1} = r_{t+j+1} \mid S_{t+j} = s_{t+j}, A_{t+j} = a_{t+j}, S_{t+j+1} = s_{t+j+1}) = 1.$$

2. **Los estados y acciones permanecen.** Queda la cadena

$$p(R_{t+k+1} = r \mid S_t = s) = \sum_{a_t, s_{t+1}, \dots, a_{t+k}, s_{t+k+1}} \left[\prod_{j=0}^{k-1} \pi(a_{t+j} \mid s_{t+j}) p(s_{t+j+1} \mid s_{t+j}, a_{t+j}) \right] \\ \times \pi(a_{t+k} \mid s_{t+k}) p(s_{t+k+1} \mid s_{t+k}, a_{t+k}) p(R_{t+k+1} = r \mid s_{t+k}, a_{t+k}, s_{t+k+1}).$$

5 Sustituir en la esperanza

$$\mathbb{E}[R_{t+k+1} \mid S_t = s] = \sum_{a_t, s_{t+1}, \dots, a_{t+k}, s_{t+k+1}} \left[\prod_{j=0}^{k-1} \pi(a_{t+j} \mid s_{t+j}) p(s_{t+j+1} \mid s_{t+j}, a_{t+j}) \right] \\ \times \pi(a_{t+k} \mid s_{t+k}) p(s_{t+k+1} \mid s_{t+k}, a_{t+k}) \bar{R}(s_{t+k}, a_{t+k}, s_{t+k+1}),$$

donde

$$\bar{R}(s_{t+k}, a_{t+k}, s_{t+k+1}) = \sum_r r p(R_{t+k+1} = r \mid s_{t+k}, a_{t+k}, s_{t+k+1}).$$

6 Primeros términos explícitos

$k = 0$.

$$\mathbb{E}[R_{t+1} \mid S_t = s] = \sum_{a_t, s_{t+1}} \pi(a_t \mid s) p(s_{t+1} \mid s, a_t) \bar{R}(s, a_t, s_{t+1}).$$

$k = 1$.

$$\mathbb{E}[R_{t+2} \mid S_t = s] = \sum_{a_t, s_{t+1}, a_{t+1}, s_{t+2}} \pi(a_t \mid s) p(s_{t+1} \mid s, a_t) \\ \times \pi(a_{t+1} \mid s_{t+1}) p(s_{t+2} \mid s_{t+1}, a_{t+1}) \bar{R}(s_{t+1}, a_{t+1}, s_{t+2}).$$

$k = 2$.

$$\mathbb{E}[R_{t+3} \mid S_t = s] = \sum_{a_t, s_{t+1}, a_{t+1}, s_{t+2}, a_{t+2}, s_{t+3}} \pi(a_t \mid s) p(s_{t+1} \mid s, a_t) \\ \times \pi(a_{t+1} \mid s_{t+1}) p(s_{t+2} \mid s_{t+1}, a_{t+1}) \pi(a_{t+2} \mid s_{t+2}) p(s_{t+3} \mid s_{t+2}, a_{t+2}) \bar{R}(s_{t+2}, a_{t+2}, s_{t+3}).$$

7 Ensamblar el retorno

$$v_\pi(s) = \mathbb{E}[G_t \mid S_t = s] = \sum_{k=0}^{\infty} \gamma^k \mathbb{E}[R_{t+k+1} \mid S_t = s],$$

es decir,

$$v_\pi(s) = \sum_{k=0}^{\infty} \gamma^k \sum_{a_t, s_{t+1}, \dots, a_{t+k}, s_{t+k+1}} \left[\prod_{j=0}^{k-1} \pi(a_{t+j} \mid s_{t+j}) p(s_{t+j+1} \mid s_{t+j}, a_{t+j}) \right] \\ \times \pi(a_{t+k} \mid s_{t+k}) p(s_{t+k+1} \mid s_{t+k}, a_{t+k}) \bar{R}(s_{t+k}, a_{t+k}, s_{t+k+1}).$$

8 Recuperar la ecuación de Bellman

Separamos $k = 0$:

$$v_\pi(s) = \underbrace{\sum_{a_t, s_{t+1}} \pi(a_t | s) p(s_{t+1} | s, a_t) \bar{R}(s, a_t, s_{t+1})}_{k=0} + \sum_{k \geq 1} \gamma^k \sum_{a_t, s_{t+1}, \dots, a_{t+k}, s_{t+k+1}} \left[\prod_{j=0}^{k-1} \pi(a_{t+j} | s_{t+j}) p(s_{t+j+1} | s_{t+j}, a_{t+j}) \right] \times \pi(a_{t+k} | s_{t+k}) p(s_{t+k+1} | s_{t+k}, a_{t+k}) \bar{R}(s_{t+k}, a_{t+k}, s_{t+k+1}).$$

Para $k \geq 1$, separamos la primera transición (A_t, S_{t+1}) . La suma interior es $v_\pi(s_{t+1})$:

$$\sum_{k \geq 1} \gamma^k \sum_{a_t, s_{t+1}} \pi(a_t | s) p(s_{t+1} | s, a_t) v_\pi(s_{t+1}).$$

Combinando,

$$v_\pi(s) = \sum_{a_t, s_{t+1}} \pi(a_t | s) p(s_{t+1} | s, a_t) \left[\bar{R}(s, a_t, s_{t+1}) + \gamma v_\pi(s_{t+1}) \right].$$

Renombrando $a_t \rightarrow a$ y $s_{t+1} \rightarrow s'$,

$$v_\pi(s) = \sum_a \pi(a | s) \sum_{s'} p(s' | s, a) \left[\bar{R}(s, a, s') + \gamma v_\pi(s') \right].$$

9 Convención de notación

Símbolo	Significado
$S_t, A_t, R_{t+1}, \dots$	variables aleatorias (mayúsculas)
$s, a, r, s', a', \dots$	valores concretos (minúsculas)
s_{t+j}, a_{t+j}	valores de S_{t+j}, A_{t+j} en la suma
$\pi(a s)$	$p(A_{t+j} = a S_{t+j} = s)$
$p(s' s, a)$	$p(S_{t+j+1} = s' S_{t+j} = s, A_{t+j} = a)$
$\bar{R}(s, a, s')$	$\sum_r r p(R_{t+j+1} = r S_{t+j} = s, A_{t+j} = a, S_{t+j+1} = s')$